

Computer Vision Practical Report:

EDGE DETECTION USING SOBEL MATRIX ALONG X AXIS

1. AIM

To detect vertical edges in an image using the Sobel operator along the X-axis in OpenCV, display the results, and save the output image.

2. PROCEDURE

- Load the image and convert it to grayscale.
- Apply Gaussian blur to reduce noise.
- Compute horizontal gradients using `cv2.Sobel()` with `dx=1` and `dy=0`.
- Convert the gradient result to 8-bit format using `cv2.convertScaleAbs()`.
- Display the original and edge images using Matplotlib, and save the result using `cv2.imwrite()`.

3. PROGRAM CODE

```
import cv2

image = cv2.imread(r"C:\Users\adminuser\Desktop\Python CV\E17-IMG.png")

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

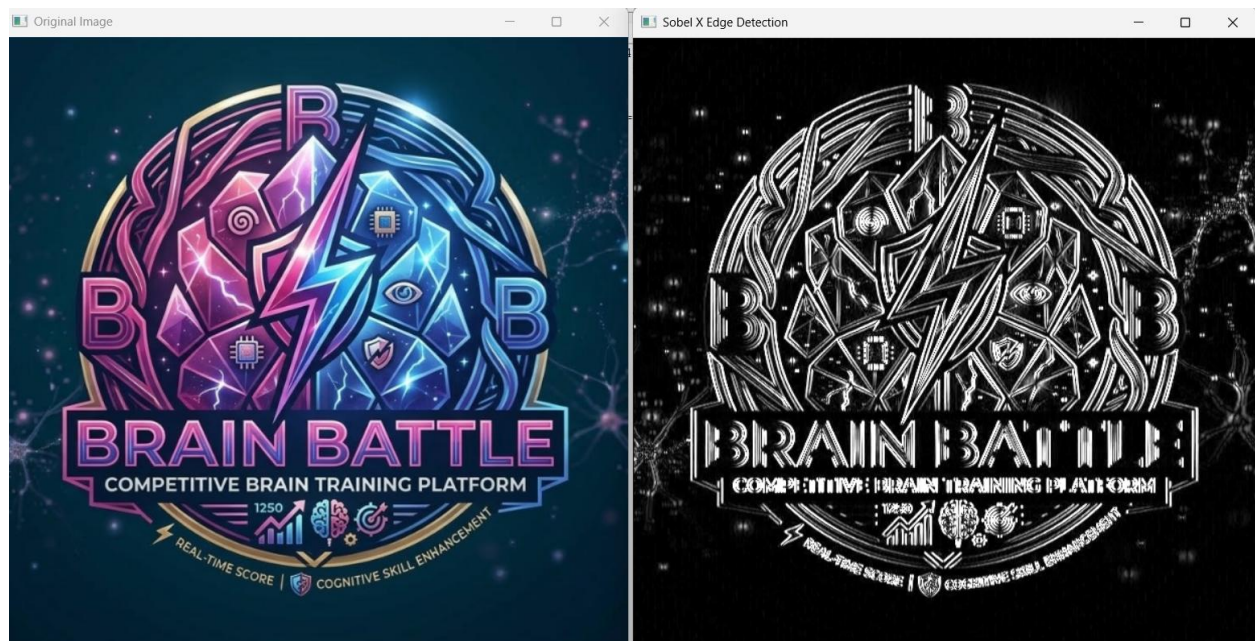
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_x = cv2.convertScaleAbs(sobel_x)

cv2.imshow("Original Image", image)
cv2.imshow("Sobel X Edge Detection", sobel_x)

cv2.imwrite("sobel_x.jpg", sobel_x)
```

```
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.